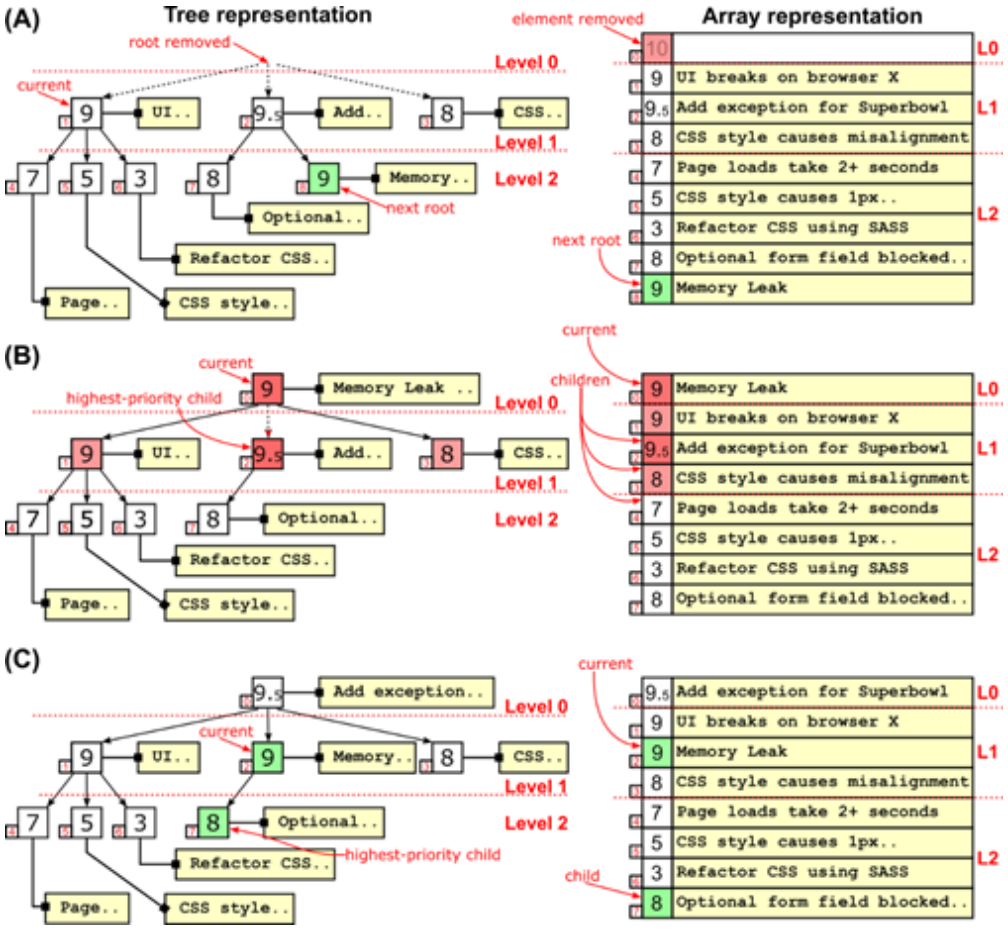


Figure 2.10 Removing the top element from a binary heap.

A) From a valid ternary heap we remove the root of the heap (storing it in a temporary location, in order to return it at the end).

B) We replace the old root with the last element in the array (which is also removed from its tail). The new root might violate heap's properties (and it does, in this example), so we need to push it down, comparing it to its children to check that the heap's properties are not violated.

C) We move the new root down, towards the heap's leaves, till we find a valid spot for it (i.e., the first one that does not violate the heap's properties).